

THE ASSOCIATION OF MATHEMATICS TEACHERS OF INDIA

Screening Test - Kaprekar Contest

NMTC at SUB-JUNIOR LEVEL {VII & VIII Standards}

MODEL PAPER

Note:

- 1) Fill in the response sheet with your Name, Class, the institution through which you appear in the specified places.
- 2) Diagrams are only visual aids; they are not drawn to scale.
- 3) You are free to do rough work on separate sheets.
- 4) Duration of the test: 2 p.m. to 4. p.m. - 2 hours.

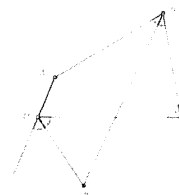
PART – A

Note:

- Only one of the choices A, B, C, D is correct for each question. Shade that alphabet of your choice in the response sheet. (If you have any doubt in the method of answering, seek the guidance of your supervisor).
- For each correct response you get 1 mark; for each incorrect response you lose $\frac{1}{2}$ mark.

SUB-JUNIOR MODEL PAPER

1. The fraction $\frac{4}{37}$ is written in the decimal form $0.a_1a_2a_3\dots$. The value of a_{2017} is
A. 8 B. 0 C. 1 D. 5
2. In the adjoining figure the breadth of the rectangle is 10 units. Two semicircles are drawn on the breadth as diameter. The area of the shaded region is 100 sq units. The shortest distance between the semicircles is
A. $\frac{5\pi}{2}$ B. $\frac{5\pi}{3}$ C. $\frac{3\pi}{4}$ D. 5π
3. If 7 rasagullas are distributed to each boy of a group, 10 rasagullas would be left. If 8 are given to each boy then 5 rasagullas would be left. So, the person who distributes the rasagullas brought 15 more rasagullas and distributed the same number (x) rasagullas to each. There is no rasagulla left. Then x is
A. 10 B. 11 C. 12 D. 14
4. Shanti has three daughters. The average age of them is 15 years. Their ages are in the ratio 3 : 5 : 7. The age of the youngest daughter is (in years)
A. 8 B. 9 C. 10 D. 12
5. In the adjoining figure, ABCD is a quadrilateral. The bisectors of $\angle B$ and the exterior angle at D meet at P. Given $\angle C = 80^\circ$; $\angle ADC = \frac{1}{2}\angle A$ and $\angle A = \angle C + 40^\circ$. Then $\angle DPB$ is



- A. 50^0 B. 60^0 C. 70^0 D. 80^0
6. The units digit of $5^{2018} - 3^{2018}$ is
A. 5 B. 6 C. 7 D. 4
7. The length of two sides of an isosceles triangle are 8 cm and 14 cm. The perimeter of the triangle (in cm) is
A. 30 B. 36 C. 19 D. 30 or 36
8. Which of the following geometric figures is possible to construct?
A. A pentagon with 4 right angled vertices.
B. An octagon with all 8 sides equal and 4 angles each of measure 60^0 and other four angles of measure 210^0 .
C. A parallelogram with 3 vertices of obtuse angle measures.
D. A hexagon with 4 reflex angles
9. Let x , y , and z be positive real numbers and let $x \geq y \geq z$ so that $x + y + z = 20.1$. Which of the following statements is true?
A. Always $xy < 99$ B. Always $xy > 1$ C. Always $xy \neq 75$ D. Always $yz \neq 49$
10. A sequence $\{a_n\}$ is generated by the rule, $a_n = a_{n-1} - a_{n-2}$ for $n \geq 3$. Given $a_1 = 2$ and $a_2 = 4$, the sum of the first 2019 terms of the sequence is given by
A) 8 B) 2692 C) -2692 D) -8

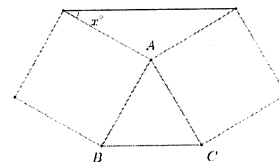
PART – B

Note:

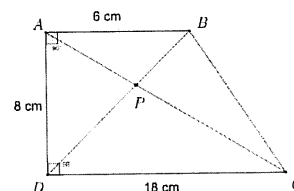
- Write the correct answer in the space provided in the response sheet.
- For each correct response you get 1 mark; for each incorrect response you lose $\frac{1}{4}$ mark.

11. Two cogged wheels of which one has 16 cogs and the other 27 cogs, mesh into each other. If the latter turns 80 times in three quarters of a minute, the number of turns made by the other in 8 seconds is _____

12. ABC is an equilateral triangle. Squares are described on the sides AB and AC as shown. The value of x is _____



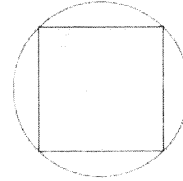
13. ABCD is a trapezium with $AB = 6$ cm, $AD = 8$ cm, and $CD = 18$ cm. The sides AB and CD are parallel, and AD is perpendicular to AB . P is the point of intersection of AC and BD . The difference between the areas of the triangles PCD and PAB in square cm is _____



14. The greatest two-digit number that divides 398, 436 and 542 leaving respectively 7, 11 and 15 as remainders is _____

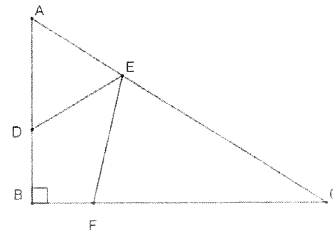
15. If $a : b = 3 : 4$ and $\frac{p}{q} = \frac{a^2 + b^2 + ab}{a^2 + b^2 - ab}$ where p, q have no common divisors other than 1, $p + q$ is _____

16. In the adjoining figure, the side of the square is $\sqrt{\frac{2018}{\pi}}$ cm. The area of the unshaded region is $\left(\frac{\pi - 2}{\pi}\right)A$ sq. cm. The value of A is _____



17. Using only the digits 1, 2, 4, 5, two-digit numbers are formed. The digits of the two-digit numbers may be the same or different. The number of such two-digit numbers is _____.

18. In the given figure, $\triangle ABC$ is a right angled triangle with $\angle ABC = 90^\circ$. D, E, F are points on AB, AC, BC respectively such that $AD = AE$ and $CE = CF$. Then, $\angle DEF =$ _____ (in degrees).



19. The number of natural numbers $n \leq 2019$ such that $\sqrt[3]{48n}$ is an integer is _____.

20. The number of *perfect cubes* that lie between $2^9 + 1$ and $2^{18} + 1$ is _____.
